AIGP Module 4 Advanced Practice Test

Governing AI Development

30 Medium to Hard Questions (Exam Simulation)

Question 1

According to Module 4, what are the four core phases of the AI system development lifecycle?

A. Plan, Design, Develop, Implement
B. Create, Test, Deploy, Monitor
C. Build, Train, Validate, Release
D. Ideate, Prototype, Launch, Maintain

Correct Answer: A

Explanation:

- **Why A is correct:** The training explicitly states: "The development of AI projects will follow the same life cycle as used for software development: plan, design, develop (or build) and implement (or deploy)." These four phases form the structured approach to AI development with governance applied at each stage.
- **Why B is incorrect:** While these activities occur during AI development, they don't represent the formal four-phase lifecycle model described in the training. Testing and monitoring are activities within the phases, not phases themselves.
- **Why C is incorrect:** These are activities that occur within the lifecycle phases but aren't the four phases themselves. Building happens in the Develop phase, training/validation happen across Design and Develop phases.
- **Why D is incorrect:** While these terms are used in some development methodologies, the training specifically identifies Plan, Design, Develop, and Implement as the four core phases for AI development lifecycle.

Question 2

According to the Three Lines of Defense (3LOD) model applied to AI development, which group constitutes the second line of defense?

A. Management and process owners implementing risk policies
B. Teams responsible for compliance and oversight identifying emerging risks in daily operations
C. Internal audit teams performing independent audits
D. External regulators conducting compliance reviews

Correct Answer: B

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Explanation:

- **Why B is correct:** The training explicitly states: "The second line of defense is teams responsible for identifying and addressing emerging risks in daily operations by way of compliance and oversight." This line provides ongoing monitoring and risk identification separate from operational management.
- **Why A is incorrect:** This describes the first line of defense: "The first line of defense is management and process owners, who are responsible for implementing risk management policies and procedures."
- **Why C is incorrect:** This describes the third line of defense: "The third line of defense is internal staff, such as an Internal Audit team, who perform independent audits on the effectiveness of the organization's risk management efforts."
- **Why D is incorrect:** External regulators aren't part of the Three Lines of Defense model, which focuses on internal organizational governance structures. Regulators operate outside this framework.

Question 3

An AI model performs successfully when analyzing standard customer profiles but fails when encountering unusual or atypical cases. According to Module 4, what AI-specific performance attribute does this scenario illustrate?

A. Hallucinations
B. Brittleness
C. Model extraction
D. Data poisoning

Correct Answer: B

Explanation:

- **Why B is correct:** The training defines brittleness as: "performing successfully in one instance yet failing in another instance." This describes exactly the scenario—the model works for standard cases but fails on atypical ones, demonstrating lack of robustness and generalizability.
- **Why A is incorrect:** Hallucinations are "instances where a gen AI model creates content that contradicts the source or creates factually incorrect output under the appearance of fact." The scenario describes failure on atypical cases, not creating false information.
- **Why C is incorrect:** Model extraction is a security threat where "Extracting model parameters or intellectual property." The scenario describes performance variability, not a security attack.
- **Why D is incorrect:** Data poisoning involves "Tampering with training data to corrupt output." The scenario describes inherent model limitations, not malicious data corruption.

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Question 4

According to the training, what is the purpose of the "Effective Challenge Principle" used with the Three Lines of Defense model?

A. To prevent any criticism of AI models
B. To allow individuals with proper expertise to challenge risk management models to identify limitations and create more effective models
C. To eliminate the need for risk management
D. To assign blame when models fail

Correct Answer: B

Explanation:

- **Why B is correct:** The training states: "The Three lines of defense model can be used with a principle known as the Effective challenge principle. Asserts that individuals with the proper expertise should have an opportunity to challenge a risk management model to help identify its limitations and ultimately create a more effective model." This promotes continuous improvement through constructive expert critique.
- **Why A is incorrect:** The principle encourages challenge and critique, not preventing it. The purpose is to improve models through expert questioning.
- **Why C is incorrect:** The Effective Challenge Principle enhances risk management by identifying limitations, not eliminating the need for it. It's a mechanism to strengthen risk management.
- **Why D is incorrect:** The principle focuses on improving models and identifying limitations constructively, not assigning blame. It's about learning and enhancement, not fault-finding for punishment.

Question 5

A data governance team is evaluating whether they have appropriate data for an AI project. According to the "Five V's" framework, which "V" addresses the question: "How trustworthy and accurate is the data?"

A. Volume
B. Velocity
C. Veracity
D. Value

Correct Answer: C

Explanation:

- **Why C is correct:** The training explicitly defines Veracity as: "How accurate is it? How trustworthy is it?" This V specifically addresses data quality, accuracy, and trustworthiness—core concerns for AI training data.
- **Why A is incorrect:** Volume addresses "How much data do you have?" which is about quantity, not trustworthiness or accuracy.

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- **Why B is incorrect:** Velocity addresses "How often does it get updated?" which is about frequency and timeliness, not accuracy or trustworthiness.
- **Why D is incorrect:** Value addresses "What is the outcome that you want from the use of the AI system? Will the data get you there? Is it the right data to use?" This is about utility and fitness for purpose, not accuracy or trustworthiness.

Question 6

According to Module 4, what is the correct order of steps in the risk mitigation hierarchy?

A. Offset, Remediate, Minimize, Avoid
B. Avoid, Minimize, Remediate, Offset
C. Minimize, Avoid, Offset, Remediate
D. Remediate, Minimize, Avoid, Offset

Correct Answer: B

Explanation:

- **Why B is correct:** The training states: "Risk mitigation hierarchy: an approach used across different settings and industries to manage risks and their impacts. Typically, the hierarchy involves: avoiding a risk, minimizing a risk, remediating a risk's impact and offsetting a risk's impact." This order reflects prioritization—first try to avoid risks entirely, then minimize what can't be avoided, then remediate impacts, and finally offset remaining impacts.
- **Why A is incorrect:** This reverses the correct order. The hierarchy starts with avoidance (the strongest mitigation) and moves to offsetting (the weakest), not vice versa.
- **Why C is incorrect:** Avoiding risks is the first and most effective step, not minimize. The hierarchy proceeds from strongest to weakest mitigation measures.
- **Why D is incorrect:** This places remediation first, but the hierarchy begins with avoidance as the most effective risk management approach, followed by minimization.

Question 7

An organization discovers that attackers have manipulated the input data at inference time to fool the AI model into producing incorrect outputs. According to Module 4, what type of AI-specific security threat is this?

A. Model inversion
B. Model extraction
C. Data poisoning
D. Evasion

Correct Answer: D

Explanation:

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- **Why D is correct:** The training defines evasion as: "Manipulating input data to fool the model at inference time." The scenario describes exactly this—attackers manipulating inputs during operation (inference) to produce incorrect outputs.
- **Why A is incorrect:** Model inversion involves "Attackers attempt to reconstruct input data from outputs." The scenario describes manipulating inputs, not reconstructing them from outputs.
- **Why B is incorrect:** Model extraction involves "Extracting model parameters or intellectual property." The scenario describes input manipulation, not stealing model information.
- **Why C is incorrect:** Data poisoning involves "Tampering with training data to corrupt output." The scenario occurs at inference time with live inputs, not during training data preparation.

Question 8

According to Module 4, what is the PRIMARY purpose of feature engineering in AI development?

A. To increase data storage requirements
B. To improve model performance, reduce computational costs, and boost model explainability
C. To make models more complex and harder to understand
D. To eliminate the need for testing

Correct Answer: B

Explanation:

- **Why B is correct:** The training explicitly states feature engineering's purpose: "1. Improving model performance... 2. Reducing computational costs... 3. Boosting model explainability." These three benefits make feature engineering a crucial step in responsible AI development.
- **Why A is incorrect:** Feature engineering aims to create useful representations from data, which typically reduces unnecessary data storage and processing, not increases it.
- **Why C is incorrect:** The opposite is true—feature engineering aims to boost explainability and create more efficient models, not increase complexity unnecessarily.
- **Why D is incorrect:** Feature engineering is part of model development but doesn't eliminate the need for testing. The training emphasizes testing remains essential regardless of feature engineering.

Question 9

A governance team wants to track where data originated, how it changed over time, and its final destination. According to Module 4, what concept does this describe?

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A. Data encryption
B. Data lineage
C. Data minimization
D. Data velocity

Correct Answer: B

Explanation:

- **Why B is correct:** The training defines data lineage as: "Data lineage tracks the flow of data over time, providing a picture of where it originated, how it changed and its destination. It provides a record of data throughout its lifecycle." This exactly matches what the governance team wants to track.
- **Why A is incorrect:** Data encryption protects data confidentiality through encoding but doesn't track its origin, changes, or destination over time.
- **Why C is incorrect:** Data minimization is a privacy principle about collecting only necessary data, not tracking data's flow and transformations over time.
- **Why D is incorrect:** Data velocity (one of the Five V's) refers to "How often does it get updated?" which is about update frequency, not tracking data's journey through systems.

Question 10

According to Module 4, what is the relationship between data lineage and data provenance?

A. They are completely unrelated concepts
B. Data provenance specifically refers to the first instance of data or its source, while data lineage tracks the entire flow over time
C. Data lineage only tracks the source, while provenance tracks the destination
D. They are identical concepts with no difference

Correct Answer: B

Explanation:

- **Why B is correct:** The training states: "Data lineage tracks the flow of data over time, providing a picture of where it originated, how it changed and its destination" and "Data provenance is similar to data lineage but refers specifically to the first instance of the data or its source." Provenance is about origin; lineage is about the entire journey.
- **Why A is incorrect:** They are closely related concepts, both dealing with data tracking. Provenance is actually a component or specific aspect of lineage.
- **Why C is incorrect:** This reverses the correct definitions. Lineage tracks the full flow (including source and destination), while provenance specifically identifies the source.
- **Why D is incorrect:** While similar, they have distinct focuses. Provenance is specifically about origin, while lineage encompasses the entire data journey including transformations.

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Question 11

An organization is conducting pre-deployment pilots for an AI system. According to Module 4, what is the PRIMARY purpose of these pilots?

A. To increase the cost of AI development
B. To determine whether the AI works as expected and provide a chance to make updates before full deployment
C. To delay deployment indefinitely
D. To eliminate all testing requirements

Correct Answer: B

Explanation:

- **Why B is correct:** The training states: "Pre-deployment pilots: a trial phase that happens before deploying the AI. Ideally, the pilot settings and conditions will match that of production as closely as possible. Pilots are intended to determine whether the AI works as expected and provide a chance to make updates before deployment." This validates functionality in near-production conditions before full rollout.
- **Why A is incorrect:** While pilots do add cost and time, their purpose isn't to increase costs—it's to validate functionality and identify issues before broader deployment, ultimately reducing risk and potential costs from failures.
- **Why C is incorrect:** Pilots are time-limited validation phases, not indefinite delays. They're designed to validate readiness for deployment, not prevent it.
- **Why D is incorrect:** Pilots complement testing, not replace it. The training emphasizes comprehensive testing throughout development, with pilots serving as final validation in production-like conditions.

Question 12

According to Module 4, what is a "challenger model" and what is its purpose?

A. A model that competes in the marketplace
B. A new model tested against the existing model to test for drift, unexpected results, and performance changes
C. A model used only for marketing purposes
D. A model that always replaces the existing model

Correct Answer: B

Explanation:

- **Why B is correct:** The training states: "Create a 'challenger model' (a new model) to test and compare against the existing model ('champion model') to test for drift, unexpected results, etc." Challenger models provide independent validation and help detect when existing models degrade or produce unexpected outcomes.

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- **Why A is incorrect:** While the term "challenger" suggests competition, this isn't about market competition—it's about technical validation and comparison within an organization's AI governance.
- **Why C is incorrect:** Challenger models serve technical governance and validation purposes, not marketing. They're part of monitoring and quality assurance processes.
- **Why D is incorrect:** Challengers don't automatically replace existing models. They're tested against the champion model, and replacement only occurs if the challenger demonstrates superior performance or addresses issues with the champion.

Question 13

A generative AI model produces a detailed, confident-sounding answer that appears factual but actually contradicts source information and contains fabricated details. According to Module 4, what is this phenomenon called?

A. Brittleness
B. Hallucinations
C. Model inversion
D. Data poisoning

Correct Answer: B

Explanation:

- **Why B is correct:** The training defines hallucinations as: "instances where a gen AI model creates content that contradicts the source or creates factually incorrect output under the appearance of fact." This exactly describes the scenario—plausible but false information presented confidently.
- **Why A is incorrect:** Brittleness refers to "performing successfully in one instance yet failing in another instance"—inconsistent performance across cases, not creating false information.
- **Why C is incorrect:** Model inversion is a security attack where "Attackers attempt to reconstruct input data from outputs," not the model creating false information.
- **Why D is incorrect:** Data poisoning involves "Tampering with training data to corrupt output," which is an external attack during training, not the model's inherent tendency to generate plausible falsehoods.

Question 14

According to Module 4, what are the recommended best practices for selecting and using features in AI model development?

A. Use different features for training and testing to maximize variety
B. Work with subject matter experts to select features, use the same features for training and testing, avoid unnecessary features, and consider using feature flags

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C. Include all possible features regardless of relevance
D. Never involve subject matter experts in feature selection

Correct Answer: B

Explanation:

- **Why B is correct:** The training explicitly states: "Work with subject matter experts to select the features... Use the same features for training and testing... Avoid any unnecessary features that you do not need. Consider using feature flags to make it easier to address areas like the need to rollback features if an issue occurs or when deploying to multiple jurisdictions with differing requirements." This comprehensive approach ensures relevant, consistent, and manageable feature sets.
- **Why A is incorrect:** Using different features for training and testing would create invalid evaluation—the model would be tested on features it wasn't trained on. Consistency is essential for valid testing.
- **Why C is incorrect:** Including all possible features creates noise, increases computational costs, reduces explainability, and can lead to overfitting. The training emphasizes avoiding unnecessary features.
- **Why D is incorrect:** The training explicitly recommends working "with subject matter experts to select the features." Domain expertise is crucial for identifying relevant features.

Question 15

According to Module 4, operational controls should designate who owns what responsibility. Which of the following is specifically mentioned as an operational control responsibility?

A. Only marketing the AI system
B. Owning the 'kill switch' for the AI, establishing feedback and appeal mechanisms, and conducting routine audits
C. Only handling customer complaints
D. Only managing the budget

Correct Answer: B

Explanation:

- **Why B is correct:** The training states operational controls must determine who will: "Have real-time operational responsibility, Conduct intermittent or routine audits and reviews, Establish and respond to feedback and appeal mechanisms, Elevate issues when there are emergent (or emergency) situations of concern, Own the 'kill switch' for the AI." These are comprehensive governance responsibilities.
- **Why A is incorrect:** Marketing isn't mentioned as an operational control responsibility. Operational controls focus on governance, oversight, risk management, and emergency response capabilities.

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- **Why C is incorrect:** While feedback mechanisms (which could include complaints) are mentioned, operational controls encompass much more than just customer complaints—they include audits, oversight, and emergency shutdown capabilities.
- **Why D is incorrect:** Budget management isn't listed as an operational control responsibility in this context. Operational controls focus on real-time operations, auditing, feedback, emergency response, and kill switch authority.

Question 16

According to Module 4, what should the "kill switch" capability enable in an AI system?

A. Automatic system updates
B. The ability for a human to shut down an algorithm if it is not performing properly, especially for high-risk systems
C. Faster processing speeds
D. Increased data collection

Correct Answer: B

Explanation:

- **Why B is correct:** The training states: "There should be the ability for a human to shut down an algorithm if it is not performing properly, especially if it has privacy impacts or the risks of using the algorithm are so great that there cannot be any room for error." The kill switch is a critical safety mechanism for human override in emergencies.
- **Why A is incorrect:** A kill switch is about emergency shutdown capability, not automatic updates. Updates are separate maintenance activities not related to emergency shutdown.
- **Why C is incorrect:** Kill switches are safety mechanisms for deactivation, not performance optimizations for speed increases.
- **Why D is incorrect:** Kill switches are about stopping systems, not enabling data collection. They're emergency shutdown mechanisms, not data gathering tools.

Question 17

When prioritizing AI use cases, Module 4 recommends evaluating three factors. What are these three factors?

A. Cost, Speed, Accuracy
B. Impact, Effort, Fit
C. Size, Complexity, Duration
D. Revenue, Profit, Market Share

Correct Answer: B

Explanation:

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- **Why B is correct:** The training states: "Prioritize based on Impact, Effort, Fit." These three factors help organizations systematically evaluate which AI use cases to pursue first—assessing potential impact, required effort, and alignment with organizational capabilities and objectives.
- **Why A is incorrect:** While cost and accuracy are considerations in AI development, the training specifically identifies Impact, Effort, and Fit as the three prioritization factors for use cases.
- **Why C is incorrect:** These might be project management considerations, but the training specifically identifies Impact, Effort, and Fit as the use case prioritization framework.
- **Why D is incorrect:** These are business outcomes that might contribute to "Impact," but the training's prioritization framework is Impact, Effort, and Fit—broader categories than just financial metrics.

Question 18

According to Module 4, what types of testing should be performed on AI systems? (Select the MOST complete answer)

A. Only accuracy testing
B. Accuracy, robustness, reliability, privacy, interpretability, safety, and bias testing
C. Only security testing
D. Only performance testing

Correct Answer: B

Explanation:

- **Why B is correct:** The training explicitly lists: "Types of testing can include: Accuracy, Robustness, Reliability, Privacy… Interpretability, Safety, Bias." This comprehensive testing approach addresses technical performance, security, fairness, and transparency requirements.
- **Why A is incorrect:** Accuracy alone is insufficient. The training emphasizes multiple testing dimensions including security, fairness, privacy, and safety—not just accuracy.
- **Why C is incorrect:** Security is important but only one dimension. The training lists seven types of testing, with security implicit in robustness and reliability testing.
- **Why D is incorrect:** Performance testing is too narrow. The training's testing framework encompasses accuracy, fairness, privacy, safety, robustness, reliability, and interpretability—not just performance metrics.

Question 19

An organization is documenting its AI system. According to Module 4, model cards or fact sheets should be included in documentation. What information should these provide?

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A. Only the names of developers
B. Standardized information about the model including its function, output, intended use, and limitations
C. Only financial costs
D. Only marketing materials

Correct Answer: B

Explanation:

- **Why B is correct:** The training states documentation should include: "Model cards or facts sheets - Provide standardized information about the model and its function/output." The purpose is to communicate essential information about capabilities, intended use, and limitations in a standardized format supporting transparency and proper use.
- **Why A is incorrect:** While developer information might be included for accountability, model cards primarily document the model's function, capabilities, limitations, and intended use—not just personnel information.
- **Why C is incorrect:** Financial costs are business information tracked separately. Model cards document technical and functional information about the model itself, not development costs.
- **Why D is incorrect:** Model cards are technical governance documentation supporting transparency and proper use, not marketing materials designed to promote the product.

Question 20

According to Module 4, when should algorithmic impact assessments (AIAs) be performed?

A. Only after deployment when problems occur
B. In the AI model design phase, and they should be reviewed throughout the lifecycle
C. Never—they are unnecessary
D. Only for marketing purposes

Correct Answer: B

Explanation:

- **Why B is correct:** The training states: "In the AI model design phase, it is important to perform or review an impact assessment on the AI model. An impact assessment is a risk management tool used to assess an AI system's benefits, risks and limitations throughout its life cycle." This emphasizes starting early and continuing throughout the lifecycle.
- **Why A is incorrect:** Waiting until after deployment is reactive and exposes users to unassessed risks. The training emphasizes impact assessments should begin in the design phase, well before deployment.

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- **Why C is incorrect:** The training explicitly states impact assessments are important and should be performed. They're a key risk management tool, not unnecessary.
- **Why D is incorrect:** Impact assessments are governance and risk management tools, not marketing instruments. They identify risks and benefits for responsible development, not promotional purposes.

Question 21

According to Module 4, what is the recommended approach for building on existing assessment processes when conducting algorithmic impact assessments?

A. Ignore all existing assessments and start from scratch
B. Build on existing Data Protection Impact Assessments (DPIAs) or Privacy Impact Assessments (PIAs) where possible
C. Only use financial audits as a foundation
D. Existing assessments are never relevant to AI

Correct Answer: B

Explanation:

- **Why B is correct:** The training explicitly states: "Utilize existing resources and processes • Where possible, build off existing data protection impact assessments (DPIAs) or privacy impact assessments (PIAs)." This leverages established processes while adapting them for AI-specific considerations.
- **Why A is incorrect:** The training recommends building on existing processes, not starting from scratch. Leveraging existing assessments is more efficient and ensures consistency with established risk management practices.
- **Why C is incorrect:** While financial controls matter, the training specifically identifies DPIAs and PIAs as the relevant existing assessments to build upon for algorithmic impact assessments, not financial audits.
- **Why D is incorrect:** The training explicitly recommends building on existing DPIAs and PIAs, demonstrating that existing assessment processes are relevant and should be adapted for AI rather than discarded.

Question 22

A data governance team is preparing data for AI model training. According to the "Five V's," which "V" addresses the question: "Will the data help achieve the desired outcome? Is it the right data to use?"

A. Volume
B. Variety
C. Veracity
D. Value

Correct Answer: D

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Explanation:

- **Why D is correct:** The training defines Value as: "What is the outcome that you want from the use of the AI system? Will the data get you there? Is it the right data to use?" Value specifically addresses whether the data serves the intended purpose and will produce the desired outcomes.
- **Why A is incorrect:** Volume addresses "How much data do you have?" which is about quantity, not whether the data serves the intended purpose.
- **Why B is incorrect:** Variety addresses "What type of data is it? Is it structured, unstructured or another type?" which is about data formats, not whether it achieves desired outcomes.
- **Why C is incorrect:** Veracity addresses "How accurate is it? How trustworthy is it?" which is about data quality and reliability, not whether it serves the intended purpose.

Question 23

According to Module 4, what should be documented regarding the training and testing process?

A. Nothing needs to be documented
B. Only the final results
C. The processes for training and testing, including outcomes and any changes made based on testing, as compliance may require audits
D. Only the names of team members

Correct Answer: C

Explanation:

- **Why C is correct:** The training states: "It is crucial to document processes for training and testing (including outcomes and anything you changed based on testing), as compliance may require audits. Document all decisions and updates — these will be critical for informing future audits." Comprehensive documentation supports compliance, accountability, and continuous improvement.
- **Why A is incorrect:** Documentation is explicitly required. The training emphasizes documenting "all decisions and updates" throughout the development process, especially training and testing.
- **Why B is incorrect:** Documentation should include processes, decisions, changes, and rationale—not just final results. Understanding how results were achieved is essential for audits and improvements.
- **Why D is incorrect:** While team member information might be included for accountability, documentation should focus on processes, decisions, outcomes, and changes made during training and testing—not just personnel lists.

Question 24

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According to Module 4, adversarial testing and threat modeling serve what purpose in AI governance?

A. To increase development costs unnecessarily
B. To identify security threats and test the system's robustness against intentional attacks or manipulation
C. To eliminate the need for any other testing
D. To delay deployment indefinitely

Correct Answer: B

Explanation:

- **Why B is correct:** The training states: "Conduct adversarial testing and threat modeling to identify security threats." These techniques deliberately attempt to attack or manipulate the system to identify vulnerabilities before malicious actors can exploit them. This is crucial for security and robustness validation.
- **Why A is incorrect:** While adversarial testing adds effort, its purpose isn't to increase costs—it's to identify vulnerabilities proactively. Finding security issues before deployment typically reduces costs compared to post-deployment incidents.
- **Why C is incorrect:** Adversarial testing complements other testing types (accuracy, bias, privacy, etc.), not replaces them. The training lists multiple testing types as necessary.
- **Why D is incorrect:** Adversarial testing is a validation technique to ensure readiness for deployment, not a mechanism to delay it indefinitely. It helps identify and fix issues before release.

Question 25

According to Module 4, what is the purpose of conducting repeatability assessments in AI testing?

A. To increase processing time
B. To ensure the AI consistently produces similar outcomes when given the same inputs
C. To make the system slower
D. To eliminate all variability in outputs

Correct Answer: B

Explanation:

- **Why B is correct:** The training states: "Conduct repeatability assessments to ensure the AI consistently produces similar outcomes." Repeatability is essential for reliability—the system should behave predictably when given the same or similar inputs, not produce random or inconsistent results.
- **Why A is incorrect:** Repeatability assessment's purpose is ensuring consistent outputs, not increasing processing time. Processing speed is a separate performance consideration.

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- **Why C is incorrect:** Making systems slower isn't the goal. Repeatability ensures consistent behavior, which is about reliability and predictability, not processing speed.
- **Why D is incorrect:** Some variability may be acceptable or even desired (e.g., in generative AI producing diverse outputs). Repeatability ensures consistency in overall behavior and decision logic, not necessarily identical outputs for all scenarios.

Question 26

An organization is selecting algorithms for an AI system. According to Module 4, what factors should guide system architecture and model selection?

A. Only the popularity of the algorithm
B. Match algorithm/model to accuracy and interpretability needs, considering time, data quality, and organizational constraints
C. Always choose the most complex model available
D. Only the cost of the algorithm

Correct Answer: B

Explanation:

- **Why B is correct:** The training states: "Match algorithm/model to accuracy/interpretability needs. Time, data quality, and organizational constraints considered." This holistic approach balances technical requirements (accuracy, interpretability) with practical constraints (time, data, organizational capabilities).
- **Why A is incorrect:** Algorithm popularity isn't a decision criterion in the training. Selection should be based on requirements, constraints, and organizational needs, not trends or popularity.
- **Why C is incorrect:** More complex models aren't always better. The training emphasizes matching algorithms to needs—sometimes simpler, more interpretable models are more appropriate than complex "black box" models, especially for high-stakes decisions requiring explainability.
- **Why D is incorrect:** While cost is a constraint, it's only one consideration. The training emphasizes matching algorithms to accuracy and interpretability requirements while considering multiple constraints including time, data quality, and organizational capabilities.

Question 27

According to Module 4, what should be established regarding metrics and thresholds for AI systems?

A. No metrics are necessary
B. Establish measurement standards for performance and bias, create baselines/benchmarks, and implement ongoing performance monitoring

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C. Only measure financial returns
D. Metrics should only be checked once at deployment

Correct Answer: B

Explanation:

- **Why B is correct:** The training states: "Establish measurement standards for performance, bias, technical/legal thresholds. Create baselines/benchmarks, ongoing performance monitoring." This comprehensive approach ensures the system meets requirements initially and continues to perform appropriately over time.
- **Why A is incorrect:** Metrics are essential for evaluating AI performance, fairness, and compliance. The training dedicates a section to metric and threshold evaluation, emphasizing their importance.
- **Why C is incorrect:** While financial metrics matter to business, technical governance requires measuring performance, bias, safety, accuracy, and other technical and ethical dimensions—not just financial returns.
- **Why D is incorrect:** The training emphasizes "ongoing performance monitoring." One-time checking at deployment is insufficient—continuous monitoring is necessary to detect drift, degradation, or emerging issues.

Question 28

According to Module 4, what is the recommended approach regarding data formats when considering data gathering for AI?

A. Only use structured data
B. Understand and manage different data formats including structured, unstructured, and semi-structured data, as well as static and streaming data
C. Only use unstructured data
D. Data format doesn't matter

Correct Answer: B

Explanation:

- **Why B is correct:** The training discusses: "Data formats: structured/unstructured/semi-structured; static/streaming." AI systems may need to handle various data types depending on use cases, and governance must address the specific challenges of each format. Different formats require different processing, quality assurance, and governance approaches.
- **Why A is incorrect:** Many AI applications require unstructured data (images, text, audio). Limiting to structured data would exclude important use cases and data sources.
- **Why C is incorrect:** Structured data is valuable for many AI applications. Different use cases benefit from different data formats—there's no universal preference for only unstructured data.

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- **Why D is incorrect:** Data format significantly matters for processing, quality assurance, storage, and governance. The training specifically addresses different formats because they require different handling approaches.

Question 29

According to Module 4, what should happen if testing reveals issues or risks during the AI development process?

A. Ignore them and proceed with deployment
B. Identify and manage the issues/risks, potentially requiring model adjustments or retraining
C. Abandon all AI projects permanently
D. Only document issues without addressing them

Correct Answer: B

Explanation:

- **Why B is correct:** The training emphasizes: "Identify and manage issues/risks during training and testing" and discusses iterative processes: "Train, test, evaluate, retrain; iterative/fine-tuning." Finding issues during testing is normal and expected—the governance response is to address them through adjustments, retraining, or other mitigation before deployment.
- **Why A is incorrect:** Ignoring issues contradicts responsible AI governance. The training emphasizes identifying and managing risks, not ignoring them. Deploying systems with known issues exposes organizations to harm and liability.
- **Why C is incorrect:** Finding issues doesn't mean abandoning all AI. The iterative development process anticipates issues and provides mechanisms (retraining, adjustment, mitigation) to address them.
- **Why D is incorrect:** Documentation alone is insufficient. The training emphasizes "manage issues/risks," which means actively addressing them, not just recording them without remediation.

Question 30

According to Module 4, what is the relationship between the business problem definition and AI use case identification in the planning phase?

A. They are unrelated activities
B. Defining the business problem includes reviewing objectives/requirements and considering alternatives to AI, which informs use case identification focused on organizational mission and goals
C. Use cases should be identified before understanding the business problem
D. Neither activity is necessary for AI development

Correct Answer: B

Explanation:

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- **Why B is correct:** The training describes a logical flow: "Defining the business problem: Review objectives and requirements, Consider alternatives to AI" followed by "Use case identification: Focus on organizational mission, Identify goal and gaps." The business problem provides context and requirements that inform which use cases are appropriate and aligned with organizational needs.
- **Why A is incorrect:** These activities are closely related and sequential. Understanding the business problem (objectives, requirements, whether AI is appropriate) precedes and informs use case identification.
- **Why C is incorrect:** The logical order is first understand the business problem (what are we trying to solve? is AI appropriate?) then identify specific use cases aligned with that problem and organizational mission.
- **Why D is incorrect:** Both activities are explicitly described in the training as crucial first steps. The training states: "Defining the business context and AI use case is a crucial first step," emphasizing their necessity for responsible AI development.

End of Module 4 Advanced Practice Test

Scoring Guide:

- **27-30 correct:** Exceptional understanding - exam-ready
- **24-26 correct:** Strong understanding - minor review needed
- **21-23 correct:** Good foundation - review lifecycle phases and testing types
- **18-20 correct:** Adequate knowledge - focus on data governance and operational controls
- **Below 18:** Review Module 4 thoroughly, especially development lifecycle and testing frameworks

Key Concepts to Master for Module 4:

1. **AI Development Lifecycle:**
    - Four phases: Plan, Design, Develop, Implement
    - Activities and governance at each phase
2. **Three Lines of Defense (3LOD):**
    - First line: Management/process owners
    - Second line: Compliance/oversight teams
    - Third line: Internal audit
    - Effective Challenge Principle
3. **Five V's of Data:**
    - Volume, Velocity, Variety, Veracity, Value

Narahari

4. **Risk Mitigation Hierarchy:**
    - Avoid, Minimize, Remediate, Offset
5. **AI-Specific Security Threats:**
    - Model inversion, extraction, poisoning, evasion
6. **Testing Types:**
    - Accuracy, robustness, reliability, privacy, interpretability, safety, bias
7. **AI-Specific Performance Issues:**
    - Brittleness, hallucinations
8. **Data Governance:**
    - Data lineage vs. provenance
    - Feature engineering
    - Data preparation
9. **Operational Controls:**
    - Kill switches, feedback mechanisms, audits, escalation
10. **Documentation Requirements:**
    - Model cards, training/testing processes, decisions

Study Tips:

- Understand the purpose and sequence of development lifecycle phases
- Know the Three Lines of Defense roles and responsibilities
- Memorize the Five V's and what each addresses
- Understand different AI-specific threats and when they occur
- Practice distinguishing between brittleness and hallucinations
- Know when to use challenger models and pre-deployment pilots
- Understand the risk mitigation hierarchy order and rationale

Good luck with your AIGP certification!

Narahari